Landon Butler

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Research Interests

Dissertation — Scalable Attribution and Selection for Machine Learning via Spectral Methods

Topics — Interpretability, Data Valuation, Trustworthy ML, Signal Processing

Education

Exp. May 2027 — **University of California, Berkeley**
Ph.D. candidate in Electrical Engineering and Computer Science
Advisor: Kannan Ramchandran
Committee: Bin Yu, Alex Dimakis

May 2022 — **University of Pennsylvania**
M.S.E. in Data Science, B.S.E. in Systems Engineering
Thesis: Convolutional Learning on Multigraphs
Advisor: Alejandro Ribeiro

Experience

Fall 2026 — **Google DeepMind** — Student Researcher — *London, UK*
Topic: Interpretability for AI weather prediction.

Summer 2026 — **Snorkel AI** — Research Intern — *San Francisco, CA*
Topic: Data valuation for training frontier AI agents.

Summer 2025 — **Apple** — Machine Learning Intern — *San Diego, CA*
Topic: Machine learning for communication systems.

Summer 2024 — **Uber** — Software Engineering Intern — *Sunnyvale, CA*
Topic: Embedding models for search and retrieval.

Publications

NeurIPS 26 — **Spectral Feedback for Test-Time Alignment of Protein Diffusion Models**
Shai Dickman, Mert Cemri, Landon Butler, Kannan Ramchandran

IEEE BITS 26 — **Spectral Sparsity: A Unifying Framework for Scalable Model Interpretability Using Codes**
Justin Singh Kang, Darin Tsui, Yigit Efe Erginbas, Landon Butler, Amirali Aghazadeh, Kannan Ramchdran

ICML 26 — **An Odd Estimator for Shapley Values**
Fabian Fumagalli, Landon Butler, Justin Singh Kang, Kannan Ramchandran, R. Teal Witter

NeurIPS 25 (Spotlight) — **ProxySPEX: Inference-Efficient Interpretability via Sparse Feature Interactions in LLMs**
Landon Butler*, Abhineet Agarwal*, Justin Singh Kang*, Yigit Efe Erginbas, Bin Yu, Kannan Ramchandran

ICML 25 — **SPEX: Scaling Feature Interaction Explanations for LLMs**
Justin Singh Kang*, Landon Butler*, Abhineet Agarwal*, Yigit Efe Erginbas, Ramtin Pedarsani, Kannan Ramchandran, Bin Yu

NeurIPS 24 — **Learning to Understand: Identifying Interactions via the Mobius Transform**
Justin Singh Kang, Yigit Efe Erginbas, Landon Butler, Ramtin Pedarsani, Kannan Ramchandran

ICASSP 24 — **Non-Commutative Convolutional Signal Models in Neural Networks: Stability to Small Deformations**
Alejandro Parada-Mayorga, Landon Butler, and Alejandro Ribeiro

IEEE TSP 23 — **Convolutional Learning on Multigraphs**
Landon Butler, Alejandro Parada-Mayorga, and Alejandro Ribeiro

IEEE TSP 23 **Convolutional Filters and Neural Networks with Non-Commutative Algebras**
Alejandro Parada-Mayorga, Landon Butler, and Alejandro Ribeiro

ICASSP 23 **Learning with Multigraph Convolutional Filters**
Landon Butler, Alejandro Parada-Mayorga, and Alejandro Ribeiro

CEUS 23 (Best Paper) **Equitable Optimization of U.S. Airline Route Networks**
Andy Eskenazi, Arnav Joshi, Landon Butler, and Megan Ryerson

ICRAT 22 (Best Paper) **Democratizing Aviation Emissions Estimation: Development of an Open-Source, Data-Driven Methodology**
Andy Eskenazi, Landon Butler, Arnav Joshi, and Megan Ryerson

IROS 21 **Learning Connectivity for Data Distribution in Robot Teams**
Ekaterina Tolstaya*, Landon Butler*, Daniel Mox, James Paulos, Vijay Kumar, and Alejandro Ribeiro

Teaching

Fall 24 Signals and Systems, *University of California, Berkeley*

Fall 21 Graph Neural Networks, *University of Pennsylvania*

Fall 21 Foundations of Data Science, *University of Pennsylvania*

Spring 21 Statistics for Data Science, *University of Pennsylvania*

Fellowships

2022 Graduate Research Fellowship, *National Science Foundation*

2021 Littlejohn Fellowship, *University of Pennsylvania*